

Total No. of Questions : 8]

PE2473

SEAT No. :

[Total No. of Pages : 2

[6584]-454

B.E. (Computer Engineering)

MACHINE LEARNING

Honours in Artificial Intelligence and Machine Learning

(2019 Pattern) (Semester - VII) (410301)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Make suitable assumption whenever necessary.

- Q1)** a) What is curse of dimensionality? Explain SVM in detail. [8]
b) Write note on regression and kernels. [9]

OR

- Q2)** a) Explain Random Forest and overfitting in detail. [8]
b) Write note on Adaptive Nearest Neighbour? [9]

- Q3)** a) Explain Learning Boolean Function in detail. [9]
b) Explain two class & multiclass discrimination. [9]

OR

- Q4)** a) What is Learning Time & Recurrent Network? Explain in detail. [9]
b) Explain Bayesian Neural Network in detail with the help of example. [9]

- Q5)** a) What is association rule mining? Explain the Apriori algorithm with the help of example. [8]
b) Explain the Proximity Matrices in detail. [9]

OR

P.T.O.

- Q6)** a) Explain the following : [8]
- i) K-Medoids
 - ii) Hierarchical Clustering
- b) Explain the self-organizing maps in detail. [9]

- Q7)** a) Explain in detail the discrete Markov Process in detail. [9]
- b) Explain in detail the Hidden Markov Model in detail. [9]

OR

- Q8)** a) What are different graphical models? Explain any two in detail. [9]
- b) What is belief propagation? Explain Chains, Trees, polytrees, Junction Trees. [9]

